INCH-POUND

MIL-C-24643/13C
14 March 1994
SUPERSEDING
MIL-C-24643/13B
3 October 1986

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 300 VOLTS, TYPE LSTTRS (INCLUDING VARIATION LSTTRSA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

This cable can be supplied in two variations: LSTTRS (unarmored) and LSTTRSA (armored).

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor - AWG size 20, class B stranding, uncoated.
Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Cross-linked polyethylene insulation. Nominal wall thickness 0.020 inch; wall thickness may vary from nominal as necessary, in order to meet the specified electrical requirements. One black and one white or natural conductor for each pair.
Fourth - Two conductors, one black and one white or natural conductor, cabled together to form a pair. The length of lay shall be determined to permit a completed cable to meet the twisting and bending endurance tests.
Fifth - Braided shield of copper, uncoated or tin coated. Minimum coverage of 85 percent.

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Construction (non-watertight) (continued)

Sixth - Shield insulation of two polyester tapes, type G in accordance with MIL-I-631, outer tape sealed. An alternate shield insulation may consist of one tape, type G of MIL-I-631, plus an extruded jacket of a suitable transparent material, nominal thickness of the transparent material 0.003 inch. The standard identification code shall be applied by method 2 on the inner tape.

Seventh- The specified number of pairs (see table I) shall be cabled together with left hand lay. The length of lay shall be determined to permit a completed cable to meet the twisting and bending endurance tests. Cabling sequence shall be consecutive, starting with pair number 1, from center outward. A single pair shall not be used as center for the cable core. Fillers shall be used in the center and valleys, as necessary, to form a firm, rounded assembly.

Eighth - Binder tape applied helically with overlap.

Construction variant for type LSTTRS

* Ninth - Cross-linked polyolefin jacket.

Construction variant for type LSTTRSA

Ninth - Cross-linked polyolefin jacket.

* Tenth - Braided metal armor.

* TABLE I. Details

Military[1/] part no. M24643/13	Size[2/]	Number of shielded pairs	Cable wall thickness (min avg) (inches)	Overall Diameter (inches)			
				Type LSTTRS		Type LSTTRSA	
				min	max	min	max
-01UN	2	2	0.050	0.629	0.680	0.679	0.730
-02UN	4	4	.050	.685	.740	.735	.790
-03UN	6	6	.050	.814	.880	.864	.930
-04UN	8	8	.050	.916	.990	.966	1.040
-05UN	10	10	.050	.999	1.080	1.049	1.130
-06UN	12	12	.070	1.017	1.100	1.067	1.150
-07UN	16	16	.070	1.100	1.190	1.150	1.240

1/ Unarmored. When armor option is required, see MIL-C-24643 for configuration of part number.

2/ Type LSTTRS and LSTTRSA.

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EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor Resistance - ohms/1000 feet at 25°C, maximum	11.95
Voltage withstand - volts, root mean square, minimum	
conductor to conductor	2000
shield to shield	200
* Insulation resistance - megohms/1000 feet, minimum	
Conductor to shield	500
Shield to shield	100
* Conductor and shield continuity	No failure
Group A:	
Visual and dimensional	No failure
Mutual capacitance - pF/ft at 1 megahertz (MHz), maximum	28
Capacitance unbalance - percent maximum	10
Characteristic impedance - ohms, at 1 MHz minimum	75
maximum	90
* Group B:	
Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Tensile strength - lb/in^2,minimum	700
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Twisting and bending endurance - (type LSTTRS only)	(see note 1)
At 75°C, cycles, minimum	6000
At minus 30 ±2°C, cycles, minimum	2000

Note: 1. Roller diameter for bending endurance test shall be as follows:

Cable type and size	Roller diameter (inches, maximum)
LSTTRS-2	2.0
LSTTRS-4	2.2
LSTTRS-6	3.0
LSTTOP-8	3.5
LSTTOP-10	4.0
LSTTOP-12	4.0
LSTTOP-16	4.5

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EXAMINATION AND TESTS:

	Group C:	Requirements
	Physicals (aged)	
	Insulation	
	Air oven	
	Tensile strength - percent of unaged, minimum	80
	Elongation - percent of unaged minimum	80
	Jacket (cable)	
	Air oven	
	Tensile strength - percent of unaged, minimum	60
	Elongation - percent of unaged minimum	60
*	Hot oil immersion	
	Tensile strength - percent of unaged, minimum	50
	Elongation - percent of unaged minimum	50
	Shrinkage	No failure
*	Permanence of printing (jacket) - cycles, minimum	125
	Heat distortion - percent of unaged, maximum	30
*	Armor (Type LSTTRSA) - conformance to material construction and coverage	No failure
	Group D:	
*	Flame propagation (cable)	No failure

QUALIFICATION INSPECTION:

* Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Aging and compatibility (cable) (125 ±5°C.)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
* Shield covering	2
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
* Fillers	45
Insulation	45

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QUALIFICATION INSPECTION (continued):	Requirements
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:
All sizes 1500 feet nominal.

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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